

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Scala for D.S Practical

Practical - 30

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Class :SY-Bsc.IT	Batch : 00
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PRACTICAL 30

Aim:

Cluster a dataset into two groups using k-means clustering in Breeze.

Code :

```
build.sbt x Demo30.scala
1 ThisBuild / version := "0.1.0-SNAPSHOT"
2
3 ThisBuild / scalaVersion := "2.13.12"
4
5 libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
6 lazy val root = (project in file("."))
7   .settings(
8     name := "Demo30"
9   )
10
```

build.sbt

Demo30.scala
 ×

```

1      import breeze.linalg._
2      object Demo30 extends App {
3          val data = DenseVector(1.0, 2.0, 3.0, 10.0, 11.0, 12.0)
4
5          var center1 = 2.0
6          var center2 = 10.0
7
8          for (i <- 1 to 5) {
9              val cluster1 = data.toArray.filter(x =>
10                  math.abs(x - center1) < math.abs(x - center2)
11              )
12
13              val cluster2 = data.toArray.filter(x =>
14                  math.abs(x - center2) <= math.abs(x - center1)
15              )
16              center1 = cluster1.sum / cluster1.length
17              center2 = cluster2.sum / cluster2.length
18          }
19          println("Cluster 1:")
20          println(data.toArray.filter(x =>
21              math.abs(x - center1) < math.abs(x - center2)
22          ).mkString(", "))
23
24          println("Cluster 2:")
25          println(data.toArray.filter(x =>
26              math.abs(x - center2) <= math.abs(x - center1)
27          ).mkString(", "))
28
29          println("Center 1 = " + center1)
30          println("Center 2 = " + center2)
31          println("\nDivyanshu Singh S039")
32      }
--

```

Output:

Run Demo30 x

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```
"C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...  
Cluster 1:  
1.0, 2.0, 3.0  
Cluster 2:  
10.0, 11.0, 12.0  
Center 1 = 2.0  
Center 2 = 11.0  
  
Divyanshu Singh S039  
  
Process finished with exit code 0
```